

II B.E. (Information Technology)
Class Test III November 2022
Subject: ITR3C3 : Data Structures

Time: 70 min

Max Marks: 20

Note : Attempt all questions. Make and state suitable assumptions when it is required while answering. Write code with proper indentation and clarity.

- Q1** Given the access to the pointer of the root of a binary search tree, write a function to return 5
the pointer of parent node to a node having **key** value. If a node with key value does not
exist in the tree then return NULL. In case if the key is on root node, return root pointer.
bstnode ParentOf (int key)*
- Q2** Consider implementing a MIN HEAP using a single dimension array. Having access to the 5
array *elements[]* , size of array as **SIZE** and a count **NoOfElement** , write function to
insert an element in the heap. *void insert (int value)*
- Q3** Consider you are given access to the pointer of the root of a binary tree (Linked 5
Implementation). Write Code or Algorithm or Idea to test whether the tree is a full binary
tree or not.
- Q4** For a binary tree, following are its Inorder and Preorder traversal sequence: 5
Inorder : p m q e x a d c b Preorder: a x m p q e b d c
Using the traversal sequences find and draw the tree structure.